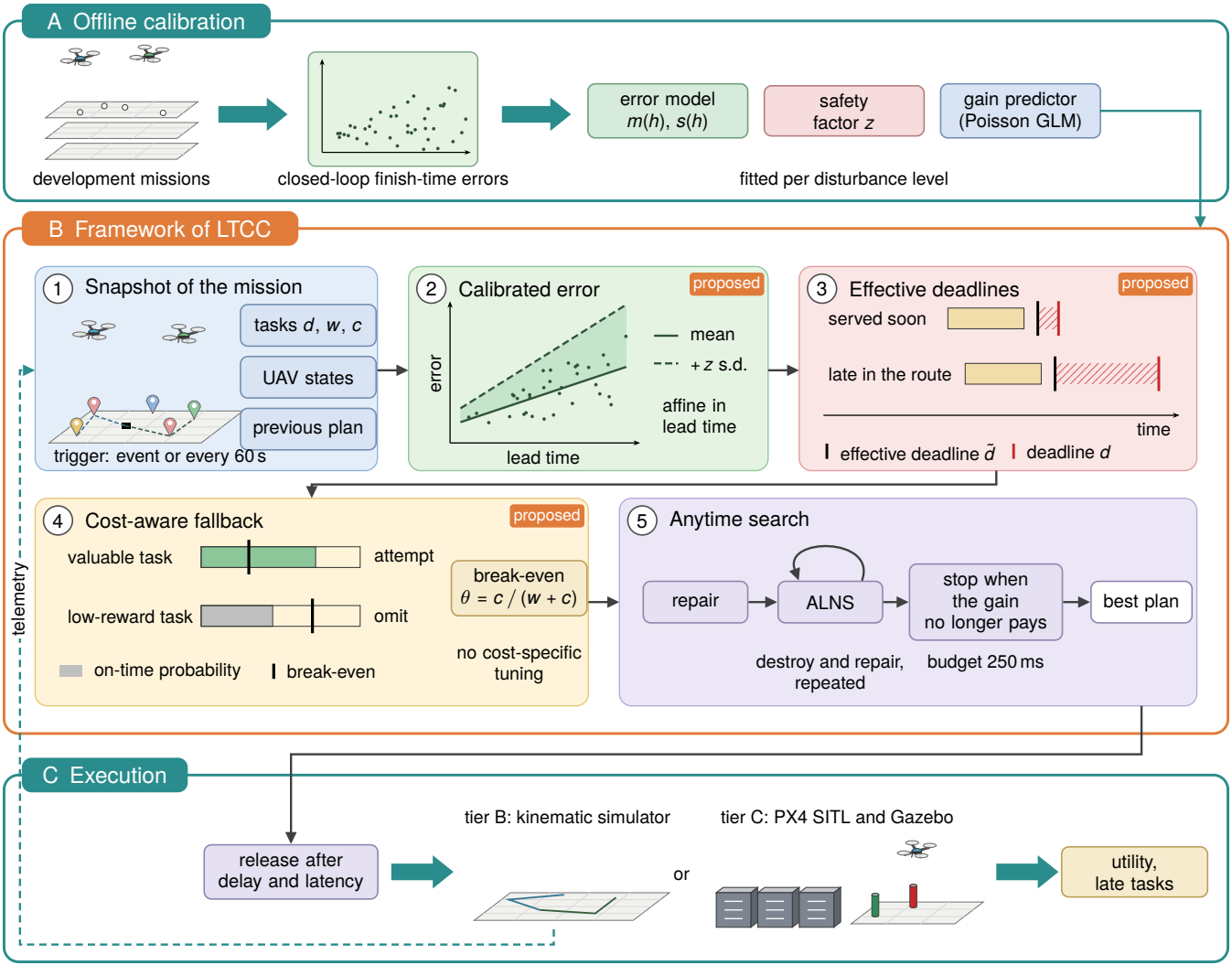

A Offline calibration
development missions
closed-loop finish-time errors
error model
m(h), s(h)
safety
factor z
gain predictor
(Poisson GLM)
fitted per disturbance level
B Framework of LTCC
1 Snapshot of the mission
tasks d, w, c
UAV states
previous plan
trigger: event or every 60 s
2 Calibrated error
proposed
error
lead time
mean
+z s.d.
affine in
lead time
3 Effective deadlines
proposed
served soon
late in the route
time
effective deadline d̃
deadline d
4 Cost-aware fallback
proposed
valuable task
attempt
low-reward task
omit
on-time probability
break-even
break-even
θ = c / (w + c)
no cost-specific
tuning
5 Anytime search
repair
ALNS
stop when
the gain
no longer pays
best plan
destroy and repair,
repeated
budget 250 ms
telemetry
C Execution
release after
delay and latency
tier B: kinematic simulator
or
tier C: PX4 SITL and Gazebo
utility,
late tasks